

A Dual-Tank Model for Real-Time Tracking of Phosphocreatine and Glycolytic Energy Reserves in Cycling

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Abstract

Endurance-cycling analytics have converged on the Critical Power (CP) / W' framework, in which all work performed above a sustainable power asymptote is drawn from a single finite reserve, W' . This lumps two physiologically distinct anaerobic systems — the fast phosphocreatine (PCr / alactic) system and the slower glycolytic (lactic) system — into one tank with a single recovery time constant. That simplification is convenient but discards the information a rider most wants during hard, variable-intensity efforts: *which* reserve is being spent, and *how fast* each will come back. This paper proposes a **dual-tank model** that splits W' into a fast PCr reserve and a slow glycolytic reserve, drives both from the instantaneous power trace, and applies system-specific depletion and restoration laws drawn from the bioenergetic and ^{31}P -MRS literature. The model is deliberately reduced to be computable at 1 Hz on a wrist- or bar-mounted device, and we specify a Garmin Connect IQ implementation that displays live reserve, consumption, and depletion for both systems. We close with a validation strategy and an explicit statement of the model's assumptions and limits.

1. Introduction

A power meter reports mechanical output, not metabolic cost. The bridge between the two is a model of the body's energy-supply systems. The dominant bridge in cycling — the CP/ W' model — treats supra-CP work as depletion of a single reservoir and sub-CP work as its exponential refill (Skiba et al. 2012). It is power-native, validated, and computationally trivial, which is why it underlies W' bal displays in GoldenCheetah, intervals.icu, and several head units.

Its central simplification is also its central weakness for real-time decision-making: **W' is one tank.**

Physiologically, anaerobic capacity is at least two tanks with very different dynamics (Margaria; di Prampero & Ferretti 1999):

- The **phosphocreatine (PCr / alactic)** system — high power, low capacity, **fast** oxidative recovery (seconds to a minute), but with recovery that is pH-sensitive and slows across repeated bouts.

- The **glycolytic (lactic)** system — larger capacity, lower peak power, and **slow** recovery (minutes) that effectively only proceeds at low intensity.

A rider deciding whether to cover the next surge cares which tank is empty. If the PCr tank is full but glycolytic is depleted, a 5-second jump is fine but a 40-second dig is not — a distinction the single-tank W'bal cannot express. The goal of this paper is a model that keeps the power-native, on-device virtues of W'bal while restoring the two-system resolution.

2. Physiological background

Phosphocreatine (alactic). ATP is buffered by the creatine-kinase reaction ($\text{PCr} + \text{ADP} \rightarrow \text{ATP} + \text{Cr}$). Intramuscular PCr ($\sim 20\text{--}25 \text{ mmol}\cdot\text{kg}^{-1}$ wet muscle) is the highest-power energy source and is drawn on instantly at any power transient. Its resynthesis is oxidative and fast: classic alactic O_2 -debt half-time $\approx 30 \text{ s}$ (Margarita); ^{31}P -MRS gives recovery time constants $\tau_{\text{PCr}} \approx 20\text{--}40 \text{ s}$, strongly **pH-dependent** (acidosis slows it) and **biphasic** under high glycolytic load (Harris et al. 1976; Yoshida et al. 2013, PMID 23662804). Recovery also **slows progressively across repeated hard bouts**.

Glycolytic (lactic). Anaerobic glycolysis converts glycogen to lactate, supplying large ATP flux for $\sim 10\text{--}2 \text{ min}$. Its "fuel level" is best tracked as accumulated muscle/blood lactate — a proxy for H^+ and metabolite accumulation. Contribution decays across repeated sprints ($\approx 40\%$ of ATP in the first of $10\times 6 \text{ s}$ efforts, $<10\%$ by the last; Sci Rep 2024, DOI 10.1038/s41598-024-78916-z). Restoration is slow (lactic O_2 -debt half-time $\approx 15 \text{ min}$) and, mechanistically, proceeds fastest at low intensity, effectively ceasing above the first lactate threshold (LT1).

The ordering that matters. At any instant, demand not met by aerobic supply is met **PCr first, then glycolytic**. This sequencing — and the very different refill rates — is exactly what the dual-tank model must reproduce.

What the two "tanks" represent — and what they don't. The "tank" is a deliberate simplification for intuition and on-device display, not a claim that either system is a literal reservoir that drains. The two sides are not even the same *kind* of quantity:

- The **PCr tank** does correspond to a real, depletable substrate store — intramuscular phosphocreatine — so "PCr empty" is close to literal.
- The **glycolytic "tank"** is better understood as **tolerance to accumulating fatigue-related metabolites** than as a fuel gauge. On the $\sim 1\text{--}4 \text{ min}$ timescale it represents, muscle glycogen is nowhere near exhausted; carbohydrate depletion is a separate, hours-long limiter this model does not address. What forces power down at the limit of a hard effort is the buildup of inorganic phosphate (Pi), H^+ (acidosis), ADP, and extracellular K^+ toward the limit of tolerance — consistent with exhaustion at CP/W' coinciding with a reproducible low-PCr / high-metabolite state. (Lactate itself is largely a fuel and a marker, not the cause of force loss.)

This is the physiological reason the two recovery laws differ: PCr resynthesizes quickly, whereas the glycolytic side recovers slowly and only at low intensity because it reflects *clearance and buffering* of accumulated byproducts, which need aerobic metabolism and time — exactly why the model tracks the glycolytic state as an accumulation variable (a muscle-lactate proxy; §4.1) and gates its recovery below LT1. Muscle fatigue is multifactorial and still debated; W' deliberately lumps it into one number for usability. In short: **the PCr tank is a fuel that depletes; the glycolytic "tank" is a byproduct bucket that fills.**

3. Prior models (and why a reduced dual-tank sits between them)

Three model families exist (full survey in `literature-review-anaerobic-models.md`):

- **Family A — CP / W' -balance** (Skiba 2012/2015; Bartram 2018; Froncioni–Clarke). Power-native and on-device-friendly, but **one lumped tank, single recovery τ** . Recovery $\tau_{W'} = 546 \cdot e^{(-0.01 \cdot D_{CP})} + 316 \text{ s}$ cannot represent fast PCr + slow glycolytic simultaneously.
- **Family B — hydraulic tank models** (Morton 1986; Weigend `three_comp_hyd` 2021). **Two explicit anaerobic vessels** (AnF = phosphagen, AnS = glycolytic) plus an aerobic vessel; out-predicts W'_{bal} for intermittent recovery (arXiv 2108.04510). But its 8 parameters are abstract and fit per-athlete by evolutionary computation — heavy for an embedded target.
- **Family C — bioenergetic supply/demand ODEs** (EJAP 2023, PMID 37369795). Explicit alactic / lactic / aerobic metabolic-rate terms, with an efficiency-limited PCr recovery and a power-gated lactate-removal law. Mechanistically ideal but has 17 parameters and needs grey-box fitting.

The gap: Family A is implementable but under-resolved; Families B and C resolve the two systems but are too heavy to run and calibrate on a head unit. The dual-tank model below is the reduction that keeps the two-tank resolution of B/C with the on-device simplicity of A, using only parameters a rider can obtain from a standard CP test plus sensible physiological defaults.

4. The dual-tank model

4.1 State and parameters

Two energy reserves (Joules), each with a fixed capacity:

- R_p — **PCr reserve**, capacity C_p
- R_g — **glycolytic reserve**, capacity C_g
- Total anaerobic capacity $W' = C_p + C_g$

Rider-supplied parameters (all obtainable from a maximal power–duration / CP test):

Symbol	Meaning	Typical default
CP	critical power (W)	from 3-/12-min CP test ($\approx$ FTP + a few %)
W'	total work above CP (J)	from same test ($\sim$ 15–25 kJ)
f_p	fraction of W' assigned to the PCr tank	0.35 (modeling choice — see §7)
P_p_max	max power the PCr tank can deliver (W)	$\approx 0.5 \cdot (\text{peak 5 s power} - \text{CP})$
τ_p	PCr recovery time constant (s)	22 (fast, pH-modulated)
τ_g	glycolytic recovery time constant (s)	360 (slow)
LT1_frac	fraction of CP below which glycolytic recovery is enabled	0.80
η	PCr recovery efficiency (hysteresis)	0.80

Derived: $C_p = f_p \cdot W'$, $C_g = (1 - f_p) \cdot W'$. Initial state $R_p = C_p$, $R_g = C_g$ (full).

4.2 The 1 Hz update

Let P be instantaneous power (W) and $\Delta t = 1 \text{ s}$. Define the demand relative to aerobic supply as $\Delta = P - CP$.

Case 1 — depletion ($\Delta > 0$, above CP). Demand $\text{need} = \Delta \cdot \Delta t$ (J) is met by **both systems in parallel**.

This matters: glycolytic ATP supply is not instantaneous. Glycogen phosphorylase (the rate-limiting glycolytic enzyme) transforms to its active form over the first several seconds of maximal effort — Parolin *et al.* (1999) measured it rising from 12% at rest to 47% by 6 s — while phosphocreatine is the immediate buffer, near-fully drawn within the first ~ 10 s (Bogdanis *et al.* 1996); both contribute from the onset, not in sequence (González-Alonso *et al.* 2000). We model this with a **glycolytic activation** term $g \in [0, 1]$ that ramps first-order with a time constant $\tau_{\text{on}} \approx 6 \text{ s}$ while above CP and relaxes back during recovery:

```

g ← g + (1 - g) · (1 - exp(-Δt/τ_on))      # activation ("glycolytic inertia"); decays b

# Capacity-proportional split (glycolytic peak rate ≈ PCr peak rate):
share_p = need / (1 + g)                    # at g=0 PCr covers all; at g=1 a 50/50 split
share_g = need - share_p

take_p = min(share_p, R_p, P_p_max·Δt)      # PCr keeps its immediate rate cap
take_g = min(share_g, R_g)
unmet = need - take_p - take_g
# any shortfall (a tank empty or rate-capped) spills to the partner, then to deficit:
take_g += min(unmet, R_g - take_g);         unmet -= ...
take_p += min(unmet, R_p - take_p, P_p_max·Δt - take_p); unmet -= ...
R_p -= take_p; R_g -= take_g

if unmet > 0: exhaustion = true              # both tanks empty/capped → rider at/over th

```

At effort onset $g \approx 0$, so PCr supplies essentially everything (the fast transient); as $g \rightarrow 1$ over a few seconds the two tanks drain together, and once the small PCr tank empties glycolysis carries the rest.

P_{p_max} is therefore the **immediate** rate ceiling, read against 1-second peak power ($P_{p_max} \approx P_{1s} - CP$). Note the split *ordering* is not identifiable from power alone — the two compartments are pinned down by the bi-exponential **recovery** of W' (§below), not by depletion; the parallel draw is the physiologically faithful choice for the live display, τ_{on} a literature-set constant rather than a fitted one.

Case 2 — restoration ($\Delta \leq 0$, at/below CP). Recovery "headroom" is $CP - P$. Each tank refills exponentially toward full; glycolytic refill is gated by intensity.

```

# PCr: fast, oxidative, efficiency-limited
R_p += η · (C_p - R_p) · (1 - exp(-Δt/τ_p))

# Glycolytic: slow, only meaningfully below LT1
if P < LT1_frac · CP:
    gate = (LT1_frac·CP - P) / (LT1_frac·CP)      # 0 at LT1, → 1 toward zero power
    R_g += gate · (C_g - R_g) · (1 - exp(-Δt/τ_g))

```

Optional realism (recommended, off by default): - *pH-slowed / fatiguing PCr recovery*: scale τ_p upward as the glycolytic tank is emptier, e.g. $\tau_{p_eff} = \tau_p \cdot (1 + k \cdot (1 - R_g/C_g))$, capturing the observed slowing across repeated bouts. - *Aerobic ramp*: replace the hard CP boundary with a first-order aerobic supply $A(t)$ that rises toward $\min(P, CP)$ with $\tau_{aer} \approx 25$ s, and use $need = (P - A) \cdot \Delta t$. This reproduces the onset "oxygen deficit" that transiently loads the anaerobic tanks even below CP.

4.3 Reported quantities

- **PCr reserve** = R_p / C_p (0–100%) — how much "punch" is left.
- **Glycolytic reserve** = R_g / C_g (0–100%) — how much "sustained dig" is left.
- **Consumption rate** (per system, W) — $take_p/\Delta t$, $take_g/\Delta t$, the live draw on each tank.
- **Combined W'bal** = $(R_p + R_g)/W'$ — backward-compatible with existing single-tank displays.
- **Depletion / exhaustion flag** — both reserves at/near zero.

4.4 Why this behaves correctly

- A short, very hard surge draws almost entirely from R_p (rate-limited glycolytic engagement), and R_p visibly refills within ~30–60 s of easy pedaling — matching PCr physiology.
- A sustained supra-CP effort empties R_p quickly, then bleeds R_g , which only comes back over minutes and only when the rider drops below LT1 — matching glycolytic physiology.
- With $f_p \rightarrow 0$ (or by summing the tanks) the model collapses to a single-tank W'bal, so it is a strict generalization of the incumbent.

5. Real-time on-device implementation (Garmin Connect IQ)

The model's per-second cost is a handful of multiplies and one `exp()` per tank — trivial for a head unit. A companion document, `connectiq-app-spec-and-prompt.md`, gives the full technical specification and a ready-to-use build prompt. Summary of the target design:

- **App type:** a custom full-screen **Data Field** (`Toybox.WatchUi.DataField`), which the head unit calls once per second — a natural fit for the $\Delta t = 1\text{ s}$ update.
- **Input:** `Activity.Info.currentPower` inside `compute(info)`.
- **Configuration:** CP , W' , f_p , and the recovery constants exposed via Connect IQ app settings (`Application.Properties`), so a rider enters them from Garmin Connect.
- **Display:** two vertical bar gauges (PCr and glycolytic reserve) with numeric % and color bands (green → amber → red), plus optional live consumption in W .
- **Recording:** write both reserves and consumption to the FIT file via `FitContributor` fields so they sync to Garmin Connect / intervals.icu / Strava for post-ride analysis.
- **Footprint:** two `Float` state variables, no arrays, well within Connect IQ memory budgets.

6. Validation strategy

The model is a hypothesis; it must be checked against data before its numbers are trusted.

1. **Face validity / unit tests** — synthetic power traces (single sprint, repeated sprints, supra-CP hold, sub-CP recovery) should produce the qualitative behaviours in §4.4.
 2. **Backward compatibility** — summing the tanks must reproduce a reference W' bal implementation (Froncioni–Clarke) on the same trace to within numerical tolerance.
 3. **Reconstitution targets** — after a full W' depletion, combined recovery should approximate the published curve ($\approx 37\%$ / 65% / 86% at 2 / 6 / 15 min; half-time ~ 234 s; Chorley & Lamb 2020).
 4. **System-specific truth (lab)** — where available, compare PCr-tank recovery against ^{31}P -MRS τ_{PCr} and glycolytic-tank state against blood-lactate kinetics.
 5. **Field calibration** — fit f_p , P_{p_max} , and τ per athlete to maximal-effort tests (e.g. a sprint-then-hold protocol) rather than trusting defaults.
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7. Assumptions and limitations

- **The PCr/glycolytic split fraction f_p is a modeling choice**, not a directly measured quantity; 0.35 is a reasonable default but should be personalized. Results are sensitive to it.
 - **Fixed time constants** ignore the documented pH-dependence and bout-to-bout slowing of PCr recovery unless the optional fatigue term is enabled.
 - **Hard CP boundary** omits the aerobic slow component and onset kinetics unless the optional aerobic-ramp term is enabled; expect over-attribution to anaerobic tanks in long low-intensity recovery (the same failure mode reported for the EJAP 2023 model).
 - **Power is whole-body and mechanical**, not muscle-specific; the model cannot see local fatigue, cadence, or fiber-type effects.
 - **Not medical or diagnostic**. This is a training/pacing aid; tank levels are estimates, not measurements of muscle chemistry.
 - **Parameters from a single CP test drift** with fitness, fatigue, heat, and altitude; periodic re-testing is required.
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8. Conclusion

The single-tank W' model earned its place by being power-native and light enough to run anywhere, but it answers "how much anaerobic work is left?" without answering "in which system?" The dual-tank model proposed here restores that resolution at negligible computational cost: split W' into a fast, rate-limited, fast-recovering PCr tank and a slow, large, slowly-recovering glycolytic tank; drain them PCr-first from the power trace; and refill them with system-specific, intensity-gated laws. It is a strict generalization of W' bal, it runs at 1 Hz on a Garmin head unit, and it produces the two numbers a rider actually races on — punch left, and dig left. The accompanying Connect IQ specification turns it into a live data field; the open questions are calibration (f_p , per-athlete τ) and field validation against system-specific lab measures.

References

See `docs/literature-review-anaerobic-models.md` for the full annotated bibliography with DOIs and PMIDs. Primary sources for this paper:

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Sourcing note: equations for the incumbent models were cross-checked against PubMed Central full text (PMC7552657, PMC10638188); typeset equations there were image-embedded, so published closed forms were reconstructed from the surviving prose and standard literature. Verify exact coefficients against publisher PDFs before hard-coding.